

GEOGRAPHY

INDIAN GEOGRAPHY



MINERALS & INDUSTRIES

MODULE - 4

- NOTES ON RATHOLE MINING
- PROBLEMS OF COAL MINING

INDIAN MINERAL & ENERGY RESOURCES

DECIDING FACTORS OF QUALITY OF COAL

1. CARBON CONTENT

- 1 Higher is the carbon content, higher is the calorific value and thus higher is the quality of the coal.
- 2 Generally, good quality coal takes a longer time for its formation.
- 3 For example, the Pennsylvanian coal of early Carboniferous times is of high quality.

2. MOISTURE CONTENT

- 1 Higher is the moisture content, lower is the calorific value and thus lower is the quality of the coal.
- 2 Good part of heat energy would be spent in evaporating water content.
- 3 Increased moisture content in coal makes the transporting to longer distance much more difficult. Coal would break up on exposure of air and mix with air forming particulate matter.

DECIDING FACTORS OF QUALITY OF COAL

3. VOLATILE MATTER



- 1 Higher the volatile matter, lower is the calorific value.
- 2 During coal burning, fumes and noise arise because of this volatile matter.
- 3 Volatile matter comprises of sulphur and ash content as well.
- 4 Majority of Indian coal has high ash content and low calorific value.

TYPES OF COAL

Based on these factors we can classify coal in to four types.

1 Swampy and marshy origin

2 Most inferior variety.

3 Contains 10 to 40 per cent carbon which implies it has more impurities.



5 Left to itself, it burns like wood, gives less heat, emits more smoke and leaves a lot of ash.

4 Contains sufficient volatile matter and lot of moisture [more smoke and more pollution].



TYPES OF COAL

1 Also known as Brown coal.

2 Lower grade coal.

3 40 to 55 per cent carbon.

4 Moisture content is high (about 30 to 35%)

5 Dark to black brown in colour

6 It undergoes quick combustion [Bad. Creates fire accidents in mines]

LIGNITE



TYPES OF COAL

BITUMINOUS COAL



1 Derives its name after a tarlike substance called bitumen.

2 60 to 80 per cent carbon.

3 Moisture and volatile content (15 to 40 per cent)
Dense, compact, and is usually of black colour.

6 This coal is therefore used in iron and steel industry also.

5 Calorific value is high due to high proportion of carbon and low moisture.

4 Does not have traces of original organic material.

TYPES OF COAL

ANTHRACITE COAL

1 Best quality; hard coal.

2 80 to 95 per cent carbon.

3 Very little volatile matter.

6 It burns with a nice short blue flame which shows there is a little or no pollutants.

5 It is difficult to ignite it but once ignited, it generates huge energy. Therefore it is highly efficient.

4 Negligibly small proportion of moisture.



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PROBLEMS OF COAL MINING

1

ENVIRONMENTAL PROBLEMS



- 1 Coal mining is associated with problems of coal dust.

2

Open cast mining ravages the whole area converting it into a rugged land.

3

Coal dust is also a problem during transportation if the power plants are not located near the mines.

4

Problems of deforestation, habitat loss, biodiversity loss and displacement of indigenous communities are also associated with coal mining.

5

The chemicals used in the dynamite contaminate the ground and surface water system.

PROBLEMS OF COAL MINING

2

COAL THEFT

Parliamentary standing committee on coal and steel in its report highlighted that **actual production of coal in India is 20% more than what is reported to the government.**



3

COAL QUALITY

- Indian coal has high ash content and low calorific value.
- This **reduces the energy output of coal** and complicates the problem of ash disposal.



PROBLEMS OF COAL MINING

4

PRODUCTIVITY



Pillar and room method used in underground coal mining.

- 1 The productivity of underground mining (output per manshift) is very low.
- 2 It is 0.8 tonnes/manshift, while the world average is around 8 tonnes/manshift.
- 3 The underground mines employ 80 percent of the man power, but contribute only 30 % of the total output.
- 4 The technique of underground mining practiced in the country – pillar and room method is wasteful and obsolete.

PROBLEMS OF COAL MINING



5

UNEVEN DISTRIBUTION OF COAL

1

The distribution of coal is highly uneven in India.

2

The major coal producing areas are confined to Jharkhand, Madhya Pradesh, Orissa and West Bengal.

3

This involves high transport cost to carry heavy commodity like coal over long distances.

4

Consequently, the coal consuming industries have to pay much higher prices for coal.

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PROBLEMS OF COAL MINING

6

LABOUR SAFETY AND HEALTH



1 The recent mishap in **Lal Matia mines in Rajmahal hills** exposed the state of safety in India coal mines.

2 **Non-compliance of standard procedures** is a major concern.

3 The government's response has always been **reactionary and not proactive**.

4 According to Directorate General of Mines Safety of the total accidental deaths in all the mines, 68% is in the coal mining.

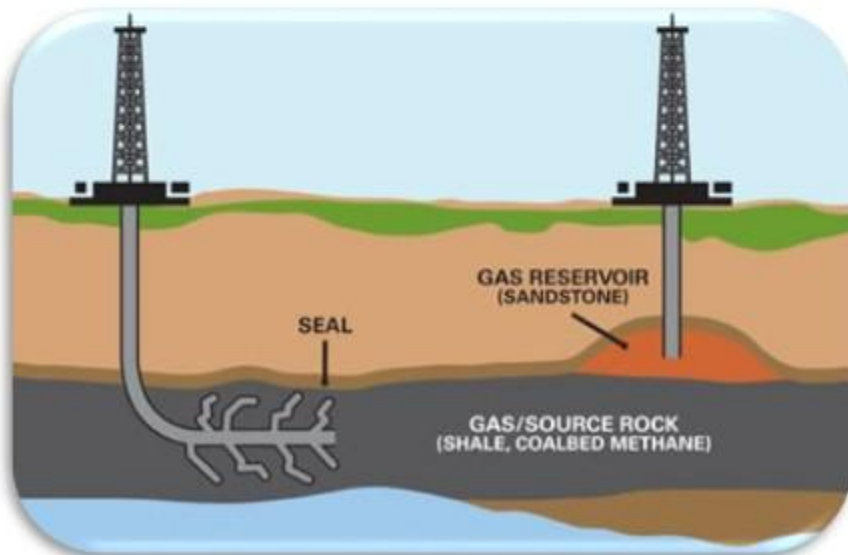
5 Coal mining is a hazardous occupation that is associated with many **respiratory diseases**.

PROBLEMS OF COAL MINING

7

INABILITY TO CAPTURE COAL BED METHANE

We shall discuss about coal bed methane in the forth coming lectures.



8

MINE CLOSURE

- Some standard procedures like land filling and sand filling are to be performed after a mine has stopped functioning and before it is shut down.
- But there have been cases where these norms are flouted.



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POINTS TO PONDER ABOUT RATHOLE MINING



WHAT IS SO CONTROVERSIAL ABOUT IT?

- 1 The National Green Tribunal (NGT) **banned it in 2014** on grounds of it being unscientific and unsafe for workers. However, the state government appealed the order in the Supreme Court.
- 2 Even after ban, it remains the **prevalent procedure for coal mining in Meghalaya** as no other method would be economically viable in Meghalaya, where the coal seam is extremely thin.

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POINTS TO PONDER ABOUT RATHOLE MINING

THEN, WHY DOES IT CONTINUE?



- 1 Political Influence:** Maximum politicians are either owners of mines or have stakes in the largely unregulated coal mining and transportation industry.
- 2 Populism:** Directly and indirectly about 2.5 lakh people are dependent on rat-hole mining economy, having influence on 16 out of 60 assembly seats.
- 3 Livelihood** of the above people in question and alternate employment opportunities are poor.
- 4 Involvement of mining mafia**

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POINTS TO PONDER ABOUT RATHOLE MINING

THEN, WHY DOES IT CONTINUE?

5

Lack of Adequate Policy:

The NGT finds The Meghalaya Mines and Mineral Policy, 2012 inadequate. The policy does not address rat-hole mining and instead states: *“Small and traditional system of mining by local people in their own land shall not be unnecessarily disturbed”*.

6

Regulation of mines is a state subject but safety of mine workers is a central subject which creates problems of implementation.

7

Lack of strict monitoring.

8

Misuse of Sixth Schedule Provisions:

- The 6th Schedule of the Constitution intends to **protect the community's ownership over its land** and the community's autonomy and consent over its nature of use.
- Coal mining currently underway in Meghalaya was a **corruption of this Constitutional Provision** wherein private individuals having private interests in earning monetary benefits from minerals vested under the land are engaging in coal mining.

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POINTS TO PONDER ABOUT RATHOLE MINING

WHO REGULATES COAL MINE SAFETY IN INDIA?

1 In India, the operations in Coalmines are regulated by the Mines Act 1952, Mine Rules of 1955, Coal Mine Regulation 1957 and several other statutes framed thereafter.

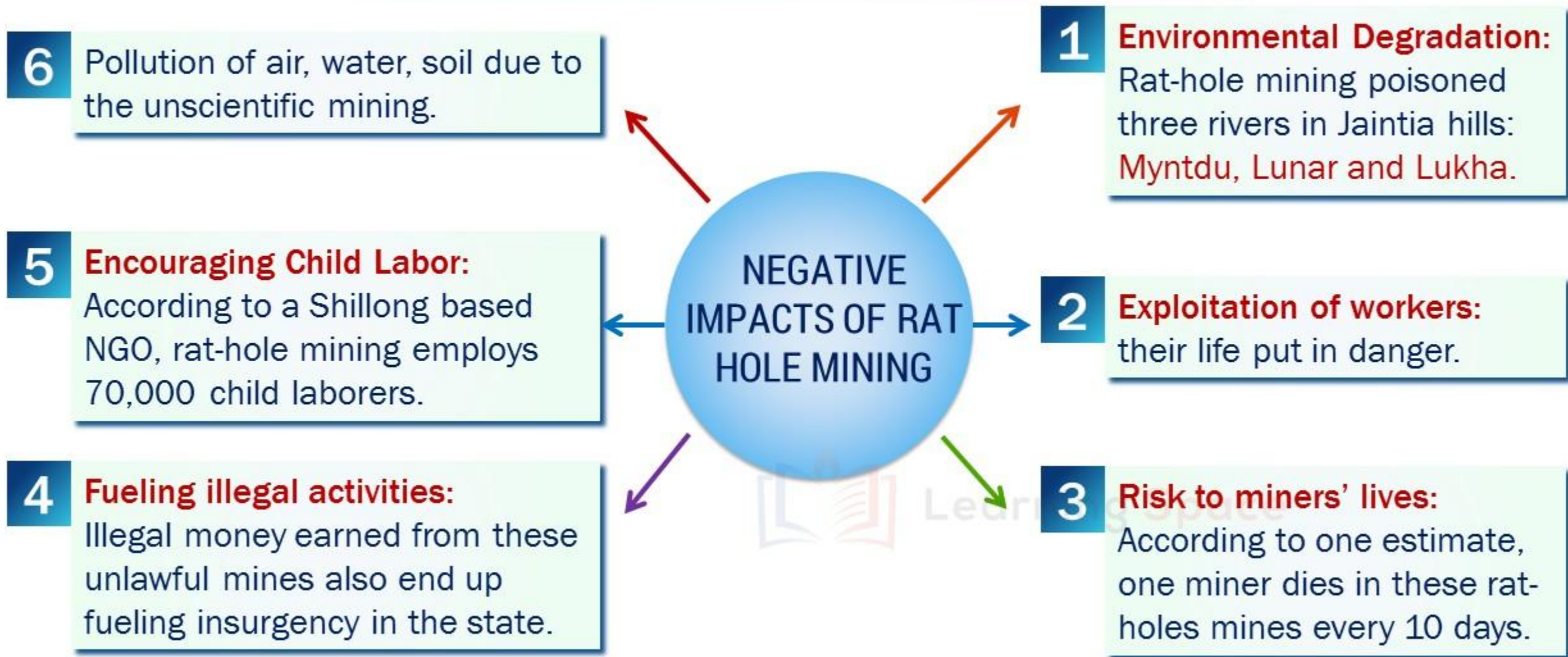


2 Directorate-General of Mines Safety (DGMS) under the Union Ministry of Labour & Employment (MOL&E) is entrusted to administer these statutes.

3 When it comes to coal mining accidents, India has a **higher proportion of deaths resulting from strata fall** (or fall of the roof and sides of underground mines) than from the use of explosives, which account for the bulk of the accidents in countries such as China and the US.

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POINTS TO PONDER ABOUT RATHOLE MINING



FODDER POINTS FOR MAINS ANSWER WRITING!!

- 6 Installation of check-posts at vulnerable points to check transport documents;
- 5 Collection of intelligence reports about illegal coal depots and illegal movement of coal and informing district authorities of the same for taking preventive action.
- 4 Fencing is being constructed at the various illegal mining sites along with displaying of signboards mentioning "Dangerous and Prohibited Place."



- 1 Rat holes created by illegal mining are being closed and filled up with stone and debris wherever possible.
- 2 Concrete walls have been erected on the mouth of the abandoned mines to prevent access and illegal activities in these areas.
- 3 Surprise raids/checks being conducted jointly by security personnel and law and order authorities of the concerned State Government.

INDIAN MINERAL & ENERGY RESOURCES

FODDER POINTS FOR MAINS ANSWER WRITING!!

STEPS BEING TAKEN TO CHECK PILFERAGE

- 1 The CVC in a statement said **coal trucks are now being fitted with GPS** to track their route, and if they divert from their scheduled path and alter their route, action is taken, making pilferage difficult.



- 2 Companies have introduced **geo-fencing of their land** to demarcate their boundary clearly thus disallowing any attempts to illegally mine valuable natural resources.

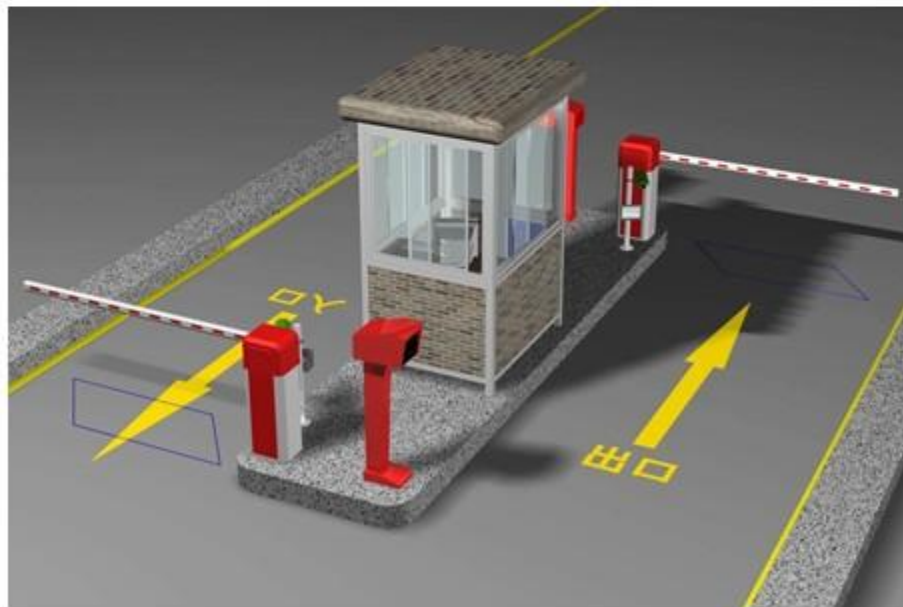


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FODDER POINTS FOR MAINS ANSWER WRITING!!

STEPS BEING TAKEN TO CHECK PILFERAGE

- 3** Radio-Frequency Identification Device (RFID) based Boom Barriers and CCTV camera at weighbridges and at all strategic location of the mines, GPRS based vehicle tracking system with geofencing, have been installed.



- 4** Regular FIRs are lodged by the Colliery Management and CISF with local Thana. A close watch on the activities of criminals is being kept by CISF.
- 5** Challans for coal transportation by trucks outside the district are being issued after fixing hologram and putting signatures of authorized officials of CISF to check pilferage.
- 6** Armed Guards have been deployed along important railway lines.
- 7** Escorting of coal rakes in coordination with RPF upto weighbridge, is arranged in pilferage prone areas.

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FODDER POINTS FOR MAINS ANSWER WRITING!!

STEPS BEING TAKEN TO CHECK PILFERAGE



- 8 Surprise re-weighment of coal loaded trucks is done at weighbridges.
- 9 Surprise checks / raids are conducted by flying squads of CISF/security department.
- 10 Joint patrolling with local police is also being carried out in pilferage prone areas.
- 11 Check posts have been established at entry / exit points where all coal laden vehicles are physically checked.
- 12 Security at coal dumps has been strengthened by fencing, proper illumination and round the clock guarding.

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